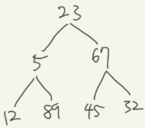
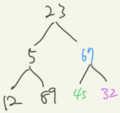
input 2 arr = 23, 5, 67, 12, 89, 45, 32

buildMaxHeap (arr)



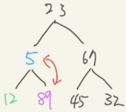
start loop (line 43) (heap.size()/2) − 1 = 2

heapify (2)



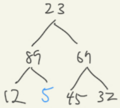
i = 2
largest = 2
left = 5
right = 6

heapify (1)



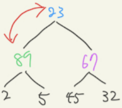
i = 1
largest = 4
left = 3
right = 4

heapify (4)



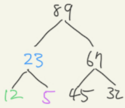
leaf

heapify (0)



i = 0
largest = 1
left = 1
right = 2

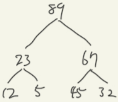
heapify (1)



i = 1
largest = 1
left = 3
right = 4

loop end

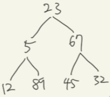
heap



output: 89 23 67 12 5 45 32

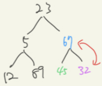
input2 arr = 23, 5, 67, 12, 89, 45, 32

buildMinHeap (arr)



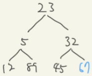
start loop (line 43) (heap.size()/2) - 1 = 2

heapify(2)



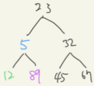
i = 2
smallest = 6
left = 5
right = 6

heapify(6)



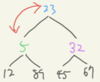
leaf

heapify(1)



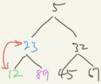
i = 1
smallest = 1
left = 3
right = 4

heapify(0)



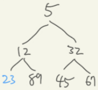
i = 0
smallest = 1
left = 1
right = 2

heapify(1)



i = 1
smallest = 3
left = 3
right = 4

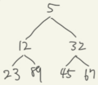
heapify(3)



leaf

loop end

heap



output: 5 12 32 23 89 45 67